

Ownership proposal across the five planners

Each planner owns a group of packing lines and everything else hangs off that: bulks go with whoever consumes them, and a purchased code only gets an owner when a single planner uses it. This document holds the split, **the rules for classifying a new code**, and why 304 purchased codes are left with no owner at all.

7,176

codes in the Ashford BOM

6,872

with a proposed owner

304

 purchased codes shared, left
unassigned

86 %

 finished goods whose whole
chain sits with one planner

The split

PLANNER	PACKING LINES	MADE	FG	BLK	WIP	CMP
Sr 1 Sr planner 1	509 Bottle Foundation Line, 513 Kalix Tube Filler, 516 Kugler, 518 IWK, 519 PKB, 521 Kalix	1,104	745	359	0	777
Sr 2 Sr planner 2	501 Kugler, 502 Kugler, 506 Kugler, 507 Kugler, 510 High Speed Kugler, 520 Romaco	1,522	879	549	94	1,097
Jr 1 Jr planner 1	306 Kemwall, 321 Book Press, 322 Book Press, 324 Book Press, 325 Vetraco Rotary Press, 326 Vetraco Rotary Press, 327 Robot Vetraco Press, 329 Camwell, 602 Robot Line, 603 Robot Line, 612 Manual Line	382	159	83	140	170
Jr 2 Jr planner 2	402 Weckerle, 410 Weckerle Tester, 416 MM360, 417 MM360	662	425	235	2	646
Intern Intern	701 Liquids Prestige	255	171	84	0	257
Unassigned — purchased codes several planners consume		0	—	—	—	304
Total		3,925	2,379	1,310	236	3,251

Made = everything with a routing in RATE, that is FG + BLK + WIP. A code can run on several machines, but it only ever has one owner.

Rules for classifying a new code

In order. The first one that fits decides, and you never need to read further down.

RULE 1

If the code is packed on a packing line, it belongs to the owner of that line.

This is the mother rule: ownership is organised by line, not by product family and not code by code. It settles 2,615 codes today, every FG and every WIP.

Every code packed on the **510 High Speed Kugler** goes to **Sr planner 2**. Every code on the **521 Kalix** goes to **Sr planner 1**. Every code on the **416 MM360** goes to **Jr planner 2**.

RULE 2

If it runs on two lines with different owners, it belongs to the owner of the smaller line.

One of the two has to be broken. The big one gets broken, because it feels it less, and the small one stays whole. Today this happens to 6 codes only.

A code running on both the **516 Kugler** (47 codes) and the **510** (262) goes to **Sr planner 1**, who owns the 516.

RULE 3

If it is a bulk, it belongs to the planner who consumes it most on their own packing lines.

You walk the whole chain up, not just to the direct parent: a bulk can feed another bulk and only show up in packing two levels higher. On a tie it goes to whoever already holds more codes on the same vessel. It settles 1,149 bulks.

A new bulk on **EKATO500ATEX** that ends up in 30 codes of the **521** and 4 of the **501** goes to **Sr planner 1**, even though both share the vessel.

RULE 4

If it is a bulk sold as it is, it belongs to whoever owns most of its vessel.

There is no packing line upstream to hang it from, so it hangs from the machine where it is made instead. It settles 161 bulks.

A bulk sold as it is, made on **BECOMIX 25L**, goes to the **Intern**, who holds 80 of its 121 codes.

RULE 5

If it is a purchased code, it belongs to the planner who consumes it — but only if there is just one.

If two or more consume it, it is left with **no owner** on purpose. It settles 2,947 purchased codes and leaves 304 unassigned.

A pack that only goes into codes of the **701** belongs to the **Intern**. A cap that goes into both the 521 and the 501 **belongs to nobody**.

The five rules cover all 7,176 codes without overlapping: 2,615 packed on a line, 1,149 bulks by consumption, 161 sold as they are, 2,947 purchased with a single owner and 304 left unassigned.

Rule 1 as a table: who owns each line

The 28 packing lines and their owner. This is the only thing you need to look up for 100 % of the finished goods.

LINE	MACHINE	CODES	OWNER	FAMILY
P05P0306	306 Kemwall	25	Jr 1 Jr planner 1	powders (pressing)
P05P0321	321 Book Press	70	Jr 1 Jr planner 1	powders (pressing)
P05P0322	322 Book Press	1	Jr 1 Jr planner 1	powders (pressing)
P05P0324	324 Book Press	49	Jr 1 Jr planner 1	powders (pressing)
P05P0325	325 Vetraco Rotary Press	37	Jr 1 Jr planner 1	powders (pressing)
P05P0326	326 Vetraco Rotary Press	32	Jr 1 Jr planner 1	powders (pressing)
P05P0327	327 Robot Vetraco Press	22	Jr 1 Jr planner 1	powders (pressing)
P05P0329	329 Camwell	11	Jr 1 Jr planner 1	powders (pressing)
P05P0402	402 Weckerle	95	Jr 2 Jr planner 2	mouldings
P05P0410	410 Weckerle Tester	57	Jr 2 Jr planner 2	mouldings
P05P0416	416 MM360	259	Jr 2 Jr planner 2	mouldings
P05P0417	417 MM360	58	Jr 2 Jr planner 2	mouldings
P05P0501	501 Kugler	356	Sr 2 Sr planner 2	liquids
P05P0502	502 Kugler	333	Sr 2 Sr planner 2	liquids
P05P0506	506 Kugler	196	Sr 2 Sr planner 2	liquids
P05P0507	507 Kugler	66	Sr 2 Sr planner 2	liquids
P05P0509	509 Bottle Foundation Line	215	Sr 1 Sr planner 1	liquids
P05P0510	510 High Speed Kugler	262	Sr 2 Sr planner 2 (256 of 262)	liquids
P05P0513	513 Kalix Tube Filler	27	Sr 1 Sr planner 1	liquids
P05P0516	516 Kugler	47	Sr 1 Sr planner 1	liquids
P05P0518	518 IWK	81	Sr 1 Sr planner 1	liquids
P05P0519	519 PKB	102	Sr 1 Sr planner 1	liquids
P05P0520	520 Romaco	79	Sr 2 Sr planner 2	liquids
P05P0521	521 Kalix	477	Sr 1 Sr planner 1	liquids
P05P0602	602 Robot Line	40	Jr 1 Jr planner 1	powders (packing)
P05P0603	603 Robot Line	20	Jr 1 Jr planner 1	powders (packing)
P05P0612	612 Manual Line	69	Jr 1 Jr planner 1	powders (packing)
P05P0701	701 Liquids Prestige	171	Intern Intern	liquids

27 of the 28 packing lines have a single owner. The only split one is the 510, because of the 6 codes it shares with the 516 that follow the 516 (rule 2).

Rule 4 as a table: who owns each vessel

You only need this for bulks sold as they are. For every other bulk rule 3 wins, and that one is about consumption, not about the vessel.

MACHINE	VESSEL	CODES	MAJORITY	SHARE
P05M0350	PORTABLEMILL	5	 Jr planner 2	100 %
P05M0351	Mill 1	5	 Jr planner 2	100 %
P05M0352	Mill 2	5	 Jr planner 2	100 %
P05M0353	Mill 3	5	 Jr planner 2	100 %
P05M0360	V250	60	 Jr planner 1	100 %
P05M0361	V5001	81	 Jr planner 1	100 %
P05M0362	V5002	81	 Jr planner 1	100 %
P05M0363	V5003	81	 Jr planner 1	100 %
P05M0364	V100 Cream	25	 Jr planner 1	100 %
P05M0453	Pan1	104	 Jr planner 2	98 %
P05M0454	Pan2	49	 Jr planner 2	100 %
P05M0455	Pan3	92	 Jr planner 2	98 %
P05M0456	Pan4	51	 Jr planner 2	100 %
P05M0457	Pan5	48	 Jr planner 2	100 %
P05M0458	Pan6	52	 Jr planner 2	100 %
P05M0460	FRYMA	1	 Jr planner 2	100 %
P05M0462	BUHLER3	45	 Jr planner 2	53 %
P05M0463	BUHLER4	45	 Jr planner 2	53 %
P05M0464	Pan8	49	 Jr planner 2	100 %
P05M0465	K8 BEAD MILL	3	 Sr planner 2	100 %
P05M0466	K60 BEAD MILL	33	 Jr planner 2	52 %
P05M0554	UNI500	10	 Jr planner 2	100 %
P05M0555	BECO150	1	 Sr planner 2	100 %
P05M0557	BECO1200	80	 Sr planner 1	70 %
P05M0562	EKATO200	187	 Sr planner 1	43 %
P05M0563	EKATO1000	75	 Sr planner 1	64 %
P05M0565	EKATO200ATEX	307	 Sr planner 2	59 %
P05M0566	EKATO320ATEX	242	 Sr planner 2	44 %
P05M0567	BECO5	92	 Sr planner 1	60 %
P05M0568	BECOMIX 2000L	280	 Sr planner 1	77 %
P05M0569	DIAF MIXER	91	 Sr planner 2	100 %

P05M0570	DIAF MIXER 2	78	Sr 2 Sr planner 2	100 %
P05M0571	DIAF MIXER 3	90	Sr 2 Sr planner 2	100 %
P05M0573	EKATO500ATEX	291	Sr 1 Sr planner 1	76 %
P05M0575	BECOMIX 25L	121	Intern Intern	66 %

Vessels are shared far more than lines: 21 of the 35 belong to a single planner, and the big liquids ones are touched by up to 4. That is normal — one vessel serves several lines — and it is exactly why ownership is organised by line and not by vessel.

Bulks: each one with whoever uses it

Of the 1,310 bulks, **1,098 are used by a single planner** and are theirs. 161 are sold as they are and no line consumes them. The **51 left are used by two or three planners at once**.

Not one is held by someone who does not use it (0 cases). The shared ones are the only place where somebody schedules against a bulk that is not theirs, and there no split can help: a bulk two planners use leaves one of them out wherever you put it. That comes to **52 hand-overs, which is exactly the floor** these lines allow. Getting below it is not done by moving bulks, it is done by moving lines.

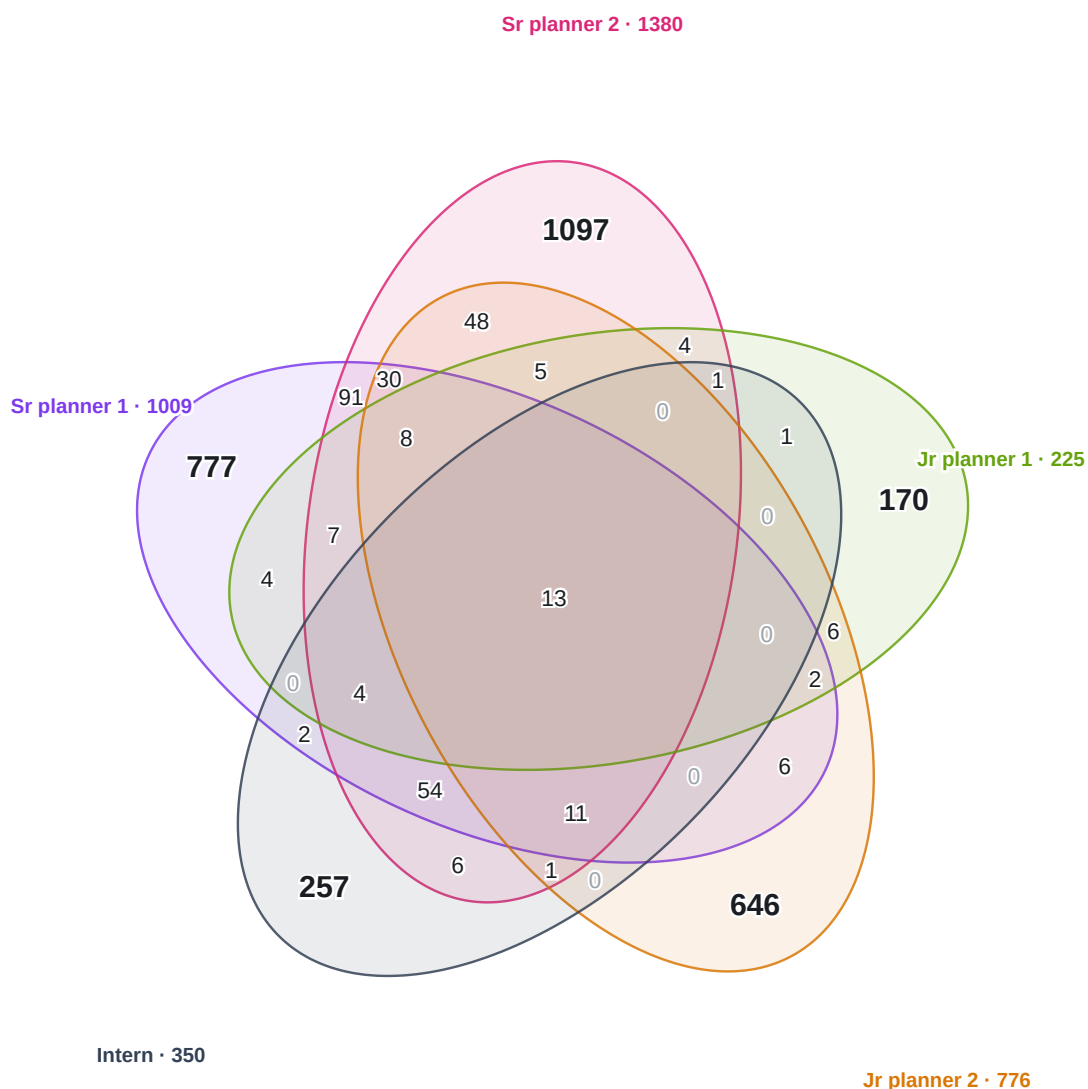
PLANNER	BULKS USED	THEIRS	SOMEONE ELSE'S
Sr 1 Sr planner 1	358	344	14
Sr 2 Sr planner 2	456	426	30
Jr 1 Jr planner 1	69	69	0
Jr 2 Jr planner 2	233	226	7
Intern Intern	85	84	1

The 304 unassigned purchased codes

Why they have no owner. They are purchased codes that go into codes belonging to two or more planners. Handing one to a single planner would be a lie: the others would keep consuming it just the same and would have to ask permission for something they use every day. In practice a shared component is not managed per planner, it is managed by purchasing, so they are deliberately left empty — and in the viewer they show up hollow, so you can see them at a glance.

They are 304 codes out of 3,251 purchased, 9 %. But they weigh far more than that suggests: **31,657 BOM lines** hang off them, because they are the packs and common materials of the whole plant.

All 3,251 purchased codes, by who consumes them



Each ellipse is one planner and holds every purchased code they consume. The number standing alone inside an ellipse is what only that planner uses, which is what they own; every number in an overlap is shared, and shared is what gets left unassigned. A zero means nobody has that exact combination.

What each planner carries

PLANNER	CONSUMED	ONLY THEIRS	SHARED	SHARE	SHARES MOST WITH
Sr 2 Sr planner 2	1,380	1,097	283	21 %	Sr planner 1 (218), Jr planner 2 (116)
Sr 1 Sr planner 1	1,009	777	232	23 %	Sr planner 2 (218), Intern (84)
Jr 2 Jr planner 2	776	646	130	17 %	Sr planner 2 (116), Sr planner 1 (70)
Intern Intern	350	257	93	27 %	Sr planner 2 (90), Sr planner 1 (84)
Jr 1 Jr planner 1	225	170	55	24 %	Sr planner 2 (42), Sr planner 1 (38)

Only theirs is what nobody else consumes, and it is exactly what they own. Shared is everything sitting in an overlap of the diagram, and none of it gets an owner.

Codes shared by each pair

PLANNER	SR 1	SR 2	JR 1	JR 2	INTERN	SHARED
Sr 1 Sr planner 1	—	218	38	70	84	232
Sr 2 Sr planner 2	218	—	42	116	90	283
Jr 1 Jr planner 1	38	42	—	34	19	55
Jr 2 Jr planner 2	70	116	34	—	25	130
Intern Intern	84	90	19	25	—	93

Each cell is how many purchased codes both planners consume, whoever else consumes them too, so a row adds up to more than its own shared total. **The two Sr planners are the heaviest pair by far, at 218 codes.** The lightest coupling in the plant is Jr planner 1 with the Intern, at 19.

How many planners share each one

CONSUMED BY	CODES	WHAT THEY ARE
2 planners	168	the normal case: two lines from different families use the same pack
3 planners	100	material common to liquids, or to liquids plus mouldings
4 planners	23	consumables used across almost the whole plant
5 planners	13	the genuinely universal ones, everybody touches them

Who shares with whom

The combinations that come up most. This is where an agreement would be needed if any of these codes ever got an owner.

SHARED BY	CODES
Sr 1 Sr 2 Sr planner 1, Sr planner 2	91
Sr 1 Sr 2 Intern Sr planner 1, Sr planner 2, Intern	54
Sr 2 Jr 2 Sr planner 2, Jr planner 2	48
Sr 1 Sr 2 Jr 2 Sr planner 1, Sr planner 2, Jr planner 2	30
Sr 1 Sr 2 Jr 1 Jr 2 Intern Sr planner 1, Sr planner 2, Jr planner 1, Jr planner 2, Intern	13

Sr 1	Sr 2	Jr 2	Intern	Sr planner 1, Sr planner 2, Jr planner 2, Intern	11
Sr 1	Sr 2	Jr 1	Jr 2	Sr planner 1, Sr planner 2, Jr planner 1, Jr planner 2	8
Sr 1	Sr 2	Jr 1		Sr planner 1, Sr planner 2, Jr planner 1	7

How many each planner appears in

PLANNER	SHARED PURCHASED CODES CONSUMED
Sr 2 Sr planner 2	283
Sr 1 Sr planner 1	232
Jr 2 Jr planner 2	130
Intern Intern	93
Jr 1 Jr planner 1	55

The column adds up to more than 304 because a code counts once for every planner that consumes it.

The ten most used

CODE	LEVEL	PARENTS	SHARED BY
99260051959	2	2,484	Sr 1 Sr 2 Jr 1 Jr 2 Intern
99400004141	2	2,473	Sr 1 Sr 2 Jr 1 Jr 2 Intern
99030037401	1	1,733	Sr 1 Sr 2 Jr 1 Jr 2 Intern
99030050096	2	1,642	Sr 1 Sr 2 Jr 1 Jr 2 Intern
99030050092	2	1,198	Sr 1 Sr 2 Jr 1 Jr 2 Intern
SHWR0200A	1	715	Sr 2 Jr 1 Jr 2
R2105	4	588	Sr 1 Sr 2 Jr 2
R1362	3	549	Sr 1 Sr 2 Jr 1 Jr 2 Intern
R1750	4	439	Sr 1 Sr 2 Jr 1 Jr 2 Intern
R3718	3	409	Sr 1 Sr 2 Intern

The first five go into more than a thousand products each and all five planners consume them. No split can assign those without leaving four people depending on a fifth. The full list, with its level and a column per planner, is in `Ashford_components_by_planner.xlsx`.

What is still open

The 510 is split

It is the only packing line with two owners. It comes from putting the 516 with foundations: 6 codes run on both and follow the 516, which is the smaller one. On top of that the bulks of the 516 feed the Kugler block far more than they feed foundations, so Sr planner 1 owns the line but schedules against bulks that Sr planner 2 largely drives. It is a known cost, accepted in exchange for levelling the load between the two Sr planners.

Kugler

It is the largest packing block in the plant and it does not come apart without breaking fluidity. While it stays whole, Sr planner 2 will carry more codes than anyone else. It is the only real lever left if the load ever has to be rebalanced.

Shared vessels

14 of the 35 vessels are touched by two or more planners. That is not a problem with the split: a vessel serves several lines by design. What has to be clear is that coordination in making is real, and that rules 3 and 4 only say who decides, not who is the only one using it.

How to adjust it

The viewer (`ashford_bom_graph_proposal.html`) already carries the split. The shape says what a code is and the colour says whose it is: a circle for an FG, a diamond for a bulk, a triangle for a WIP and a square for a purchased code; hollow means nobody holds it. Click a planner and the map keeps only their codes together with the purchased ones they share with the rest. To move things: **Select related** picks up a whole chain, the rectangle picks up an area, and **Assign to** reassigns it. Changes save themselves in the browser.

Generated by `build_report.py` from `Ashford_split_2.xlsx` and `ownership_proposal.json`. Every figure comes from the same split the viewer carries.